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where α_1 is the coefficient multiplying the T^2 piece coming from $R(e,\omega)$ (fixed by conventions and already used in Section 4.2). For $m_T^2 > 0$ this algebraic system has the unique solution

$$\boxed{T_\mu = \partial_\mu \epsilon, \qquad S^\mu = 0, \qquad q_{\lambda\mu\nu} = 0,} \tag{A63}$$

which is precisely the C1 *pure-trace alignment* (Section 4) written in covector form and implies $\mathcal{T}_\mu = 0$ on-shell.[9]

Two equivalent implementations of the *spurion limit*. We isolate the low-energy regime where observables reduce to those used in the main text:

$$\boxed{\begin{array}{r} \text{(i) Hard penalty: } m_T \to \infty \;\Rightarrow\; \mathcal{T}_\mu = 0, \\ \text{(ii) Lagrange current: } \mathcal{L}_\Lambda = \sqrt{-g}\,\Lambda^\mu \mathcal{T}_\mu. \end{array}} \tag{A64}$$

In case (i) integrating out ω (equivalently T) produces a functional delta $\delta[\mathcal{T}]$; in case (ii) the constraint is exact already at the classical level. In both realizations, keeping f_ϵ^2 finite leaves a standard $(\partial\epsilon)^2$ spectator; *the spurion posture* used in the main text corresponds to taking $f_\epsilon^2 \to 0^+$ (or treating ϵ as nondynamical) *within the two-derivative truncation*, so that observables depend only on $\partial\epsilon$ through the invariant $\mathcal{T}_\mu$ (and hence through Σ_ϵ once C1 is enforced).

Low-energy effective action (power counting). At energies $E \ll m_T$ one obtains, after eliminating ω,

$$S_{\text{eff}}^{(\text{LO})} = \int d^4x \sqrt{-g}\Big[\tfrac{M_P^2}{2} R(\mathring{\omega}) + \alpha_1\, I_T + \alpha_2\, \Sigma_\epsilon\Big] + \mathcal{O}\!\left(\frac{\partial^4}{m_T^2}\right), \qquad I_T \equiv -\tfrac{1}{4}\,\Pi_{\text{PT}}[T^A{}_{BC} T_A{}^{BC}], \tag{A65}$$

and *after* enforcing C1 (Section 4), $I_T = -6\eta^2\,\Sigma_\epsilon$, so that the LO bulk reduces to a single invariant line in the $\{I_T, \Sigma_\epsilon\}$ plane (Section 3.5).

Appendix G.2. Complementary EFT Pictures (Same Gradient-Only Physics)

Two complementary constructions emphasize that *only* $\partial\epsilon$ can enter observables at this order.

(i) Torsional–axion picture. Augment (A61) by a topological coupling

$$S_{\text{t-ax}} = \int d^4x \sqrt{-g}\,\frac{\alpha}{4}\,\epsilon\,\text{NY}, \qquad \text{NY} \equiv d\Big(e^A \wedge T_A\Big), \tag{A66}$$

which is exact. Under A5, NY is a *boundary convention*; integrating by parts and using (A63) gives bulk terms proportional to $\partial_\mu\epsilon$ contracted with improvement currents, hence no new bulk Euler–Lagrange content in the PT-even sector. The low-energy reduction again depends only on Σ_ϵ (plus boundary assignments absorbed into A5).

(ii) Axion electrodynamics–like geometry. The situation mirrors $\theta\, F\tilde{F}$ in gauge theory: the integrated density is topological, and only $\partial_\mu\theta$ is physically measurable (through Chern–Simons currents) at two derivatives. Replacing $(\theta, F\tilde{F})$ by (ϵ, NY) and using the scalar PT projector (A2) reproduces the same *gradient-only* observability for ϵ.

Appendix G.3. UV Hints and Emergence

While our analysis is agnostic about ultraviolet completions, two standard paths naturally generate a field with the required properties:

1. **Torsionful connections in string-inspired setups.** In heterotic/type II supergravities the NS–NS three-form $H = dB$ enters via the *torsionful* spin connection $\omega_\pm = \mathring{\omega} \pm \frac{1}{2} H$. In slowly varying, parity-even sectors and upon dimensional reduction, gradients of the axion-like fields dual to B

[9] Equivalently, one may add the quadratic invariant I_T with a positive coefficient and obtain the same irrep alignment. The projector statements of Section 2 ensure all scalar contractions are taken in the PT-even, real sector.

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